

A BIVARIATE t-TEST FOR SPHERICITY

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1. Introduction

Let x be a p -dimensional normal variate with mean μ and covariance matrix Σ . A random sample $x_1, \dots, x_N$ is available, and the purpose is to test the null hypothesis $H_0: \Sigma = \sigma^2 \Sigma_0$ where Σ_0 is completely specified and of rank p , and σ^2 is unknown. By replacing x_i by $\Sigma_0^{-1/2} x_i$ the null hypothesis can be reduced to $H_0: \Sigma = \sigma^2 I$, and this is the form of the null hypothesis to be considered in this paper. Write

$$x_k = (x_{1k} \dots x_{pk})', k = 1, \dots, N,$$

$$\bar{x}_j = \sum_j x_{jk} / N, j = 1, \dots, p,$$

$$\text{and } A = (a_{ij}) \quad \text{where } a_{ij} = \sum_j (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j).$$

The likelihood ratio test, derived by Mauchly (1940), is based on

$$W = |A| p^p / (\text{tr } A)^p.$$

The null distribution of W is given for $p=2$ by Mauchly (1940) and for $p=3, 4$ and 6 by Consul (1967). Gleser (1966) found the asymptotic distribution of W , which is normal. The exact distribution of W (in integral form) and critical values are given by Nagarsenker and Pillai (1972). An expression for the noncentral distribution of W in terms of zonal polynomials and G -functions is given by Khatri and Srivastava (1971).

The locally best invariant test, discussed by John (1972) and by Sugiura (1972), is based on

$$V = \text{tr } A^2 / (\text{tr } A)^2.$$

These sphericity tests can of course be applied to any Wishart matrix.

One of the main problems remaining when V or W is used for testing sphericity is that of multiple comparisons. In the present paper this problem will be discussed only for the bivariate case, $p=2$. In this case $W=2-2V$ and both tests are equivalent to the same F -test. A bivariate t -test will be derived which is directly related to this F -test.

2. A bivariate t -test

In the case $p=2$ we can write

$$(1) \quad \Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} = \sigma^2 \begin{pmatrix} 1 & \rho\tau \\ \rho\tau & \tau^2 \end{pmatrix}$$

where $\sigma^2 = \sigma_{11}$, $\rho = \sigma_{12}/(\sigma_{11} \sigma_{22})^{1/2}$ and $\tau^2 = \sigma_{22}/\sigma_{11}$.

The null hypothesis $\Sigma = \sigma^2 I$ then becomes

$$H_0 : \rho=0; \tau^2=1.$$

Write $r = a_{12}/(a_{11} a_{22})^{1/2}$ and $f = N-2$, and let

$$(2) \quad t_1 = f^{1/2} a_{12}/(a_{11} a_{22} - a_{12}^2)^{1/2} \\ = f^{1/2} r/(1-r^2)^{1/2},$$

$$(3) \quad t_2 = f^{1/2} (a_{11} - a_{22})/2(a_{11} a_{22} - a_{12}^2)^{1/2}.$$

The statistic t_1 is the usual t -statistic for testing $H_0 : \rho=0$, and its null distribution is that of Student's t with f degrees of freedom. The statistic t_2 is the corresponding t -statistic for testing $H_0 : \tau^2=1$ derived by Pitman (1939) and Morgan (1939). It is the statistic used for testing the correlation coefficient between $x_1 - x_2$ and $x_1 + x_2$, where x_1 and x_2 are the two components of x . This correlation coefficient is

$$(4) \quad \rho' = (1-\tau^2)/(1+2\tau^2+\tau^4-4\rho^2\tau^2)^{1/2},$$

and thus $H_0 : \tau^2=1$ is equivalent to $H_0 : \rho'=0$.

The joint density of t_1 and t_2 is derived as follows. The joint density of a_{11} , a_{12} and a_{22} is the Wishart distribution (Anderson (1958), p. 157). Transforming to t_1 by (1) and t_2 by (2) and $s^2 = a_{11}a_{22} - a_{12}^2$, the Jacobian being $2s^2/f \{1 + (t_1^2/f) + (t_2^2/f)\}^{1/2}$, and integrating out for s^2 from 0 to ∞ , the joint density of t_1 and t_2 is found to be

$$(5) \quad p(t_1, t_2) = \frac{(1-\rho^2)^{1/2} f^{1/2} (1-\rho'^2)^{1/2} f^{1/2}}{2\pi \Gamma(f+1) (1-\rho^2 \rho'^2)^{1/2} f^{1/2}} \sum_{i=0}^{\infty} \frac{\Gamma(f+i+1)}{i! f^{1/2} i! (1-\rho^2 \rho'^2)^{1/2} i} \\ \cdot \frac{\{t_1 \rho (1-\rho'^2)^{1/2} + t_2 \rho' (1-\rho^2)^{1/2}\}^i}{\{1 + (t_1^2/f) + (t_2^2/f)\}^{1/2} f^{1/2} i + 1}$$

Under $H_0 : \rho=0, \tau^2=1$, i.e. $\rho=\rho'=0$, the joint density of t_1 and t_2 reduces to

$$(6) \quad p(t_1, t_2) = (2\pi)^{-1} \{1 + (t_1^2/f) + (t_2^2/f)\}^{-1/2f-1}$$

which is the density of the bivariate t-distribution of Dunnett and Sobel (1954). Using critical values in Steffens (1969) and elsewhere, one can test H_0 against one- or two-sided alternatives. The Studentized maximum modulus test, which rejects the null hypothesis if $|t_1| > t^0$ or $|t_2| > t^0$ where t^0 is an appropriately chosen critical value, may be used when the alternative is $H_1: \rho \neq 0, \tau^2 \neq 1$. The power of this test is $1 - \beta_t$ where

$$\beta_t = \int_{-t^0}^{t^0} \int_{-t^0}^{t^0} p(t_1, t_2) dt_1 dt_2.$$

Transforming to polar coordinates, this integral can be written

$$(7) \quad \beta_t = \frac{(1-\rho^2)^{1/2f+1/2} (1-\rho'^2)^{1/2f+1/2}}{\pi^{1/2} \Gamma(1/2f+1/2) (1-\rho^2 \rho'^2)^{1/2f+1/2}} \int_{-\pi/4}^{\pi/4} \sum_{i=0}^{\infty} \frac{\Gamma(1/2f+i+1/2)}{\Gamma(i+1/2)(1-\rho^2 \rho'^2)^i} \\ \cdot I_{t^0}^2 / (f \cos^2 \theta + t^0{}^2)^{(i+1, 1/2f)} [(\rho(1-\rho'^2)^{1/2} \cos \theta + \rho'(1-\rho^2)^{1/2} \sin \theta)^{2i} \\ + (\rho(1-\rho'^2)^{1/2} \sin \theta + \rho'(1-\rho^2)^{1/2} \cos \theta)^{2i}] d\theta,$$

where $I_x(p, q)$ is the incomplete beta function.

This integral can be evaluated by quadrature. A table of the power of the bivariate t-test is presented in Table 2.

The marginal power of the two t-tests may be found with the aid of the tables of the noncentral distribution of the correlation coefficient by David (1938). The marginal power of t_1 depends on ρ but not on τ . The marginal power of t_2 depends on ρ' which is a function of ρ as well as τ . Even if τ is bounded away from 0 and ∞ , the power of t_2 increases to 1 as ρ^2 approaches 1. On the other hand, the marginal significance level does not depend on ρ except in the degenerate case $\rho^2 = 1$.

3. The F-Test

From the joint density (6) it follows that

$$(8) \quad F = \frac{1}{2} t_1^2 + \frac{1}{2} t_2^2$$

has, under H_0 , an F-distribution with 2 and f degrees of freedom. It is easily seen that

$$F = \frac{1}{2} f (W^{-1} - 1)$$

which is a monotonic function of W defined in section 1. The test based on F is therefore equivalent to the test based on W . The power $1-\beta_F$ of the F -test can be found by integrating the joint density (5) of t_1 and t_2 over the region

$$\frac{1}{2}t_1^2 + \frac{1}{2}t_2^2 \geq F^0$$

where F^0 is a critical value of F with 2 and f degrees of freedom. This yields the following expression which is equivalent to that obtained by Girschick (1941):

$$(9) \quad \beta_F = (1-\lambda)^{\frac{1}{2}f+\frac{1}{2}} \sum_{k=0}^{\infty} \frac{\Gamma(\frac{1}{2}f+k+\frac{1}{2})}{\Gamma(k+1)\Gamma(\frac{1}{2}f+\frac{1}{2})} \lambda^k I_{2F^0/(f+2F^0)}(k+1, \frac{1}{2}f)$$

where

$$(10) \quad \begin{aligned} \lambda &= 1 - (1-\rho^2)(1-\rho'^2)/(1-\rho^2\rho'^2) \\ &= 1 - 4\tau^2(1-\rho^2)/(1+\tau^2)^2. \end{aligned}$$

Replacing τ by $1/\tau$ or ρ by $-\rho$ does not change the value of λ . Also, $0 \leq \lambda \leq 1$ and $1-\beta_F=1$ when $\lambda=1$. Some values of the power of the F -test are given in Table 1.

The importance of (8) is that it provides a method of multiple comparisons. If the joint null hypothesis $H_0: \rho=0, \tau=1$ is rejected, one rejects independence ($\rho=0$) if $\frac{1}{2}t_1^2 > F^0$, and equality of the variances ($\tau=1$) if $\frac{1}{2}t_2^2 > F^0$.

4. Comparison of the t- and F-tests

(a) *One and two-sided testing:*

Using the bivariate t -distribution, one may perform one or two-sided tests with respect to either part of the null hypothesis. The F -test is basically two-sided in both parts.

(b) *Marginal power:*

Using the F -test, one may reject the joint null hypothesis without being able to reject any part of it. This problem does not arise with the bivariate t -test. The critical value t^0 is smaller than $(2F^0)^{1/2}$, and therefore the marginal power of the separate t -tests is larger with the bivariate t -test than with the F -test.

(c) *Invariance:*

The F -test is invariant with respect to rotation of the (x_1, x_2) -axes, while the bivariate t -test is not. Since the joint null hypothesis is invariant, the F -test is clearly preferable when testing the joint null hypothesis is the only consideration. A bivariate t -test could conceivably be used when one is rather more interested in testing the two

separate hypotheses of independence and equality of variances, while still maintaining a joint significance level, and where one wants to have the marginal power as large as possible.

(d) *Power:*

For any given value of λ , the power of the F-test does not depend on the particular values of ρ and τ . On the other hand, the power of the bivariate t-test is determined by any two of the three parameters ρ , τ and λ . One way of comparing the power of the two tests is to keep λ fixed and vary ρ (and therefore τ). When $\rho=0$, $\lambda=\rho'^2=(1-\tau^2)^2/(1+\tau^2)^2$ and when $\tau^2=1$, $\lambda=\rho^2$. Such comparisons are made in Table 3. It is seen that the bivariate t-test is more powerful when ρ is close to 0 or when τ^2 is close to 1. When both ρ and τ differ by a fair amount from their null values, the F-test is more powerful. In Table 4, λ is tabulated as a function of ρ and τ in order to facilitate comparison of Tables 1 and 2.

(e) *Confidence regions:*

A joint confidence region for ρ and τ consists of all values ρ^* and τ^* such that the null hypothesis $H_0: \rho=\rho^*, \tau=\tau^*$ is not rejected, using the F- or bivariate t-test. In the case of the F-test the confidence region is defined by

$$(11) \frac{(\tau^2 a_{11} + a_{22} - 2\rho\tau a_{12})^2}{4\tau^2(1-\rho^2)(a_{11}a_{22} - a_{12}^2)} \leq (f+2F^0)/f,$$

cf. Anderson (1958), p. 264.

Using the bivariate t-test, the confidence region is defined by two inequalities

$$(12) f^{1/2} |a_{12} - \rho\tau a_{11}| / (a_{11}a_{22} - a_{12}^2)^{1/2} \leq t^0$$

and

$$\frac{f^{1/2} |\tau^2 a_{11}(1-\rho^2) + 2\rho\tau a_{12} - a_{22}|}{2\tau(1-\rho^2)^{1/2} (a_{11}a_{22} - a_{12}^2)^{1/2}} \leq t^0$$

These two types of regions are illustrated in Figure 1 for the special case

$a_{11} = a_{12} = 1; a_{22} = 4$ based on 14 observations.

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TABLE 1

Power (in percentage) of the F-test for sphericity with noncentrality parameter λ , significance level 0,05 and f degrees of freedom.

$\lambda \backslash f$	6	12	18	24	30	40	60	120
0,05	6,8	9,3	12,0	14,8	17,7	22,6	32,6	59,7
0,10	8,9	14,5	20,6	26,9	33,2	43,3	61,1	90,3
0,15	11,3	20,7	30,5	40,2	49,3	62,6	81,3	98,5
0,20	14,1	27,6	41,2	53,6	64,2	77,7	92,4	99,8
0,30	21,1	43,6	62,5	76,5	85,8	94,3	99,3	100,0
0,40	30,2	60,6	80,4	91,0	96,1	99,1	100,0	
0,50	41,7	76,6	92,1	97,6	99,3	99,9		
0,60	55,7	89,1	97,8	99,6	99,9	100,0		
0,70	71,6	96,5	99,7	100,0	100,0			
0,80	87,0	99,5	100,0					
0,85	93,2	99,9						
0,90	97,6	100,0						
0,95	99,7							

TABLE 2

Power (in percentage) of the bivariate t-test for sphericity with significance level 0,05; noncentrality parameters ρ and τ and f degrees of freedom.

		$f=6$						
$\rho \backslash \tau$		1,0	1,5	2,0	2,5	3,0	3,5	4,0
0,0		5,0	11,4	27,7	47,6	65,1	77,8	86,2
0,2		6,4	12,8	29,3	49,4	66,7	79,1	87,2
0,4		12,1	18,3	35,1	55,2	71,8	83,1	90,0
0,6		27,7	33,0	48,7	67,2	81,1	89,5	94,3
0,7		42,9	47,1	60,5	76,2	87,3	93,4	96,6
0,8		65,1	67,6	76,6	87,1	93,7	97,1	98,6
0,9		90,2	90,9	93,7	96,9	98,7	99,5	99,8

		$f=12$						
$\rho \backslash \tau$		1,0	1,5	2,0	2,5	3,0	3,5	4,0
0,0		5,0	20,9	56,0	82,2	93,8	98,0	99,3
0,2		8,3	24,0	58,6	83,7	94,5	98,3	99,4
0,4		22,6	36,4	67,3	88,3	96,4	98,9	99,7
0,5		36,7	48,3	74,6	91,5	97,6	99,3	99,8
0,6		56,0	64,2	83,3	94,9	98,7	99,7	99,9
0,7		77,4	81,7	91,9	97,8	99,5	99,9	100,0
0,8		93,8	95,0	97,9	99,5	99,9	100,0	100,0

TABLE 2 (continued)

$\rho \backslash \tau$		f=18						
		1,00	1,25	1,50	1,75	2,00	2,25	2,50
0,0		5,0	11,7	30,9	55,6	75,9	88,5	94,9
0,2		10,4	16,8	35,3	59,1	78,3	89,8	95,6
0,4		33,6	38,4	53,0	71,4	85,7	93,7	97,5
0,5		53,9	57,2	67,6	80,8	90,8	96,2	98,5
0,6		75,9	77,6	83,2	90,4	95,6	98,3	99,4
0,7		92,4	92,9	94,7	97,1	98,7	99,5	99,8
0,8		99,1	99,2	99,4	99,7	99,9	100,0	100,0

$\rho \backslash \tau$		f=24						
		1,00	1,25	1,50	1,75	2,00	2,25	2,50
0,0		5,0	14,4	40,8	69,5	87,7	95,8	98,7
0,2		12,6	21,4	46,2	72,9	89,4	96,5	99,0
0,3		24,8	32,4	54,3	77,6	91,5	97,3	99,2
0,4		44,2	49,9	66,6	84,1	94,3	98,3	99,5
0,5		67,8	71,1	80,9	91,3	97,0	99,2	99,8
0,6		87,7	88,9	92,8	96,8	99,0	99,7	99,9
0,7		97,7	97,9	98,7	99,4	99,8	100,0	100,0

$\rho \backslash \tau$		f=30						
		1,0	1,2	1,4	1,6	1,8	2,0	2,2
0,0		5,0	12,8	35,7	63,6	83,9	94,0	98,1
0,2		14,8	21,9	42,9	68,3	86,3	95,1	98,5
0,3		30,6	36,3	53,8	74,9	89,5	96,4	98,9
0,4		53,9	57,8	69,6	83,9	93,5	97,9	99,4
0,5		78,2	80,0	85,7	92,6	97,1	99,1	99,8
0,6		94,0	94,5	96,1	98,0	99,3	99,8	99,9

$\rho \backslash \tau$		f=40						
		1,0	1,2	1,4	1,6	1,8	2,0	2,2
0,0		5,0	15,8	46,6	77,2	93,1	98,4	99,7
0,2		18,7	28,1	55,0	81,4	94,6	98,8	99,8
0,3		40,1	47,0	67,2	86,8	96,3	99,2	99,9
0,4		67,6	71,4	82,5	93,2	98,2	99,6	99,9
0,5		89,3	90,6	94,3	97,9	99,5	99,9	100,0
0,55		95,3	95,8	97,5	99,1	99,8	100,0	100,0

$\rho \backslash \tau$		f=60						
		1,0	1,1	1,2	1,3	1,4	1,5	1,6
0,00		5,0	9,1	22,2	43,1	65,1	82,0	92,1
0,15		16,3	19,9	31,6	50,2	69,7	84,6	93,3
0,20		26,9	30,0	40,3	56,7	73,8	86,8	94,3
0,25		41,0	43,6	51,9	65,2	79,1	89,6	95,6
0,30		57,2	59,1	65,2	74,9	85,0	92,6	96,9
0,35		72,9	74,1	77,9	84,1	90,6	95,4	98,1
0,40		85,5	86,1	88,2	91,5	95,0	97,6	99,0
0,50		97,8	97,9	98,2	98,7	99,3	99,7	99,9

TABLE 2 (continued)

$\rho \backslash \tau$	f=120						
	1,00	1,10	1,15	1,20	1,25	1,30	1,40
0,00	5,0	13,9	25,8	41,9	59,2	74,4	92,8
0,10	14,9	22,9	33,6	48,1	63,6	77,3	93,7
0,15	29,7	36,3	45,3	57,3	70,1	81,4	94,9
0,20	50,5	55,2	61,5	70,0	79,1	87,1	96,5
0,25	71,9	74,5	78,1	83,0	88,2	92,7	98,0
0,30	87,8	89,0	90,5	92,7	94,9	96,9	99,2
0,40	99,2	99,3	99,4	99,5	99,7	99,8	99,9

TABLE 3

Power of the F- and bivariate t-test with significance level 0,05; $\lambda = 0,25$ and f degrees of freedom.

		f=8	f=18	f=40	f=60
F-test		0,233	0,521	0,881	0,974
t-test	$\rho=0,00$	0,242	0,539	0,893	0,978
	$\rho=0,10$	0,239	0,530	0,887	0,976
	$\rho=0,20$	0,229	0,507	0,867	0,968
	$\rho=0,30$	0,218	0,476	0,839	0,957
	$\rho=0,40$	0,213	0,464	0,828	0,952
	$\rho=0,46$	0,224	0,492	0,855	0,963
	$\rho=0,48$	0,231	0,512	0,872	0,970
	$\rho=0,50$	0,242	0,539	0,893	0,978

TABLE 4

λ as a function of ρ and τ

$$\lambda = 1 - 4\tau^2(1 - \rho^2) / (1 + \tau^2)^2$$

$\rho \backslash \tau$	1,0	1,5	2,0	2,5	3,0	3,5	4,0
0,0	0,000	0,148	0,360	0,524	0,640	0,721	0,779
0,1	0,010	0,156	0,366	0,529	0,644	0,724	0,781
0,2	0,040	0,182	0,386	0,543	0,654	0,732	0,787
0,3	0,090	0,225	0,418	0,567	0,672	0,746	0,798
0,4	0,160	0,284	0,462	0,600	0,698	0,766	0,814
0,5	0,250	0,361	0,520	0,643	0,730	0,791	0,834
0,6	0,360	0,455	0,590	0,696	0,770	0,821	0,858
0,7	0,490	0,565	0,674	0,757	0,816	0,858	0,887
0,8	0,640	0,693	0,770	0,829	0,870	0,900	0,920
0,9	0,810	0,838	0,878	0,910	0,932	0,947	0,958

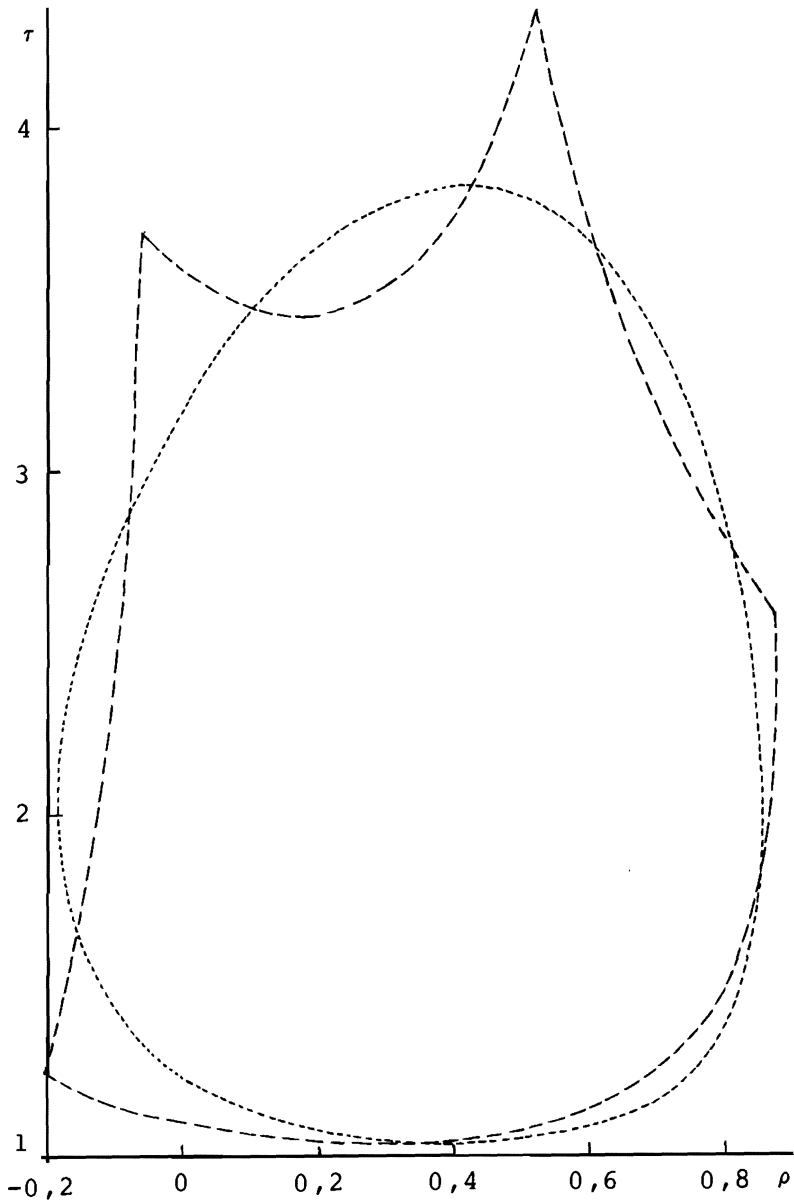


FIGURE 1 – Joint confidence region for ρ and τ based on an estimated matrix $a_{11} = a_{12} = 1$; $a_{22} = 4$ with 14 observations and significance level 0,05 using the F-test (---) and the bivariate t-test (-.-)

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